

Market Research for Production Game

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Market Overviews

Clinical

The clinical market segments are similar to the medical segments described below, except that a clinician would provide the monitor to a patient, but the patient, instead of a physician, would have primary responsibility for monitoring the data.

Law

For these markets, a court order would require an offender to wear the monitor, continually transmitting toxicology and location data to law enforcement, generally over the course of several days or weeks. Police departments and other government organizations would purchase the monitors to measure alcohol or different families of drugs in the bloodstream of individuals, mostly parolees. These markets are made up of six independent segments. In each segment, the agencies care about a different type of toxicology monitoring, but they all care about tracking the location of the monitored individuals, and the ability to receive continuous data updates over a cellular network, as opposed to Bluetooth, connection (with rough equivalence between the 2.4 GHz and 5 GHz options).

Medical

These markets consist of physicians monitoring patients who are undergoing various therapies. There are 14 independent market segments made up of different medical practices. A department or physician would purchase a monitor appropriate for their specific practice, provide it to a patient as part of that patient's treatment, and then review the collected data online. Selling a monitor to one segment would have no effect on the consumers in the other medical segments.

Feature Descriptions

Platform

The electronics and monitors are added to a wearable “platform” that keeps the monitors close to the skin. People that wear the monitors prefer smaller monitors (wrist bands) over larger monitors that would typically be worn under clothes. Some markets (e.g., general population, fashion-conscious consumers) have a stronger preference for this than others.

GPS

Adding a GPS to monitors allows the monitor results to be correlated with location. This feature is imperative for some markets (e.g., law enforcement), and nice-to-have for others.

Network

To be useful the monitor must be able to transmit data for storage on another device like a smartphone or a device on the cloud. Bluetooth transmits data to a smartphone or other hand-held device. The 2.4 GHz and 5 GHz interfaces allow the monitor to transmit data over cellular networks. As with GPS, this is a very important requirement for certain markets.

Power

The monitor must have power to work, and it is designed to run on a rechargeable battery. The nickel cadmium (Ni-Cd) battery is small but basic; it can only last a few hours on full charge. The polymer battery, while of similar size to the Ni-Cd, is lighter and can last several days on one charge. The two battery pack power options simply combine multiple batteries (of either the nickel cadmium or polymer variety, depending on the type of pack) into an arrangement that is wearable around the waist, which while rather bulky and heavy for the casual user, has the merit of increasing the time between charges by a factor of about ten.

Finish

The original monitor encapsulates the hardware in utilitarian plastic, which is functional, but not particularly appealing. The monitor could be encapsulated in dyed or painted plastic, with matching materials for the platform garment. Current customers have not requested new finishes but the sales representatives hope that a more fashionable looking monitor might drive downstream demand, particularly in athletic markets.

Clinical Fertility Market

The fertility B2C market is made up of women who are interested in continually monitoring their fertility cycles. They currently can buy a device that will monitor their Reproductive LH at a cost of \$50-\$150. Your company is capable of creating a more convenient version of this device, as well as a combined LH / FSH device that will increase the monitor's precision.

Regional Market Size

Region	Market Size
Each Region	40,000-60,000

Bass Model Parameters

Parameter	Value
p	0.00025
incremental p from every \$500 in daily advertising	0.00025
q	0.004

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Hormone)	WTP
LH	130-300
LH / FSH	230-400

Note: These WTP values assume products that contain a reasonable set of features beyond one of the core features listed in the table.

Deal Breakers

- The discomfort of wearing bulky battery packs.

Other Considerations

- The end users in this market have a slight preference for wearing devices on their wrists and rather dislike wearing them around their chests.
- Minor considerations regarding GPS connectivity and upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 10.

Law (Narcotic) Market

In this market, most of the time a court order would require an offender to wear a monitor, continually transmitting the data to law enforcement.

Regional Market Size

Region	Market Size (Narcotic)
Each Region	5,000-15,000

Bass Model Parameters

Parameter	Value
p	0.00025
incremental p from every \$500 in daily advertising	0.00025
q	0.0025

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Toxicology)	WTP
Narcotic	1,100-1,600

Note: These WTP values assume products that contain a reasonable set of features beyond the core feature listed in the table.

Deal Breakers

- Lack of GPS.
- Lack of cellular network connectivity.

Other Considerations

- Minor considerations such as the utility of various power configurations, the exact GHz bandwidth of the cellular network connection, and the appeal of upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 90.

MD Cancer (Breast) Market

This market consists of oncologists monitoring patients who are undergoing courses of therapies that can cost tens of thousands of dollars. The base panel monitors common biomarkers for most common cancers. The other monitors add specific checks for less common biomarkers and common drugs to treat those specific types of cancer.

Regional Market Size

Region	Market Size (Breast)
Each Region	10,000-20,000

Bass Model Parameters

Parameter	Value
p	0.0002
incremental p from every \$500 in daily advertising	0.0002
q	0.0035

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Cancer)	WTP
Base Panel	0-900
Breast	900-1,600

Note: These WTP values assume products that contain a reasonable set of features beyond one of the core features listed in the table.

Deal Breakers

- None.

Other Considerations

- Minor considerations such as the convenience of various power configurations, GPS connectivity, and the appeal of upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 30.

MD Fertility (Estrogen) Market

This market is made up of physicians who are interested in using your estrogen monitoring devices in lieu of sequences of blood tests costing several hundred dollars.

Regional Market Size

Region	Market Size (Estrogen)
Each Region	10,000-20,000

Bass Model Parameters

Parameter	Value
p	0.0002
incremental p from every \$500 in daily advertising	0.0002
q	0.0035

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Hormone)	WTP
Estrogen	575-965

Note: These WTP values assume products that contain a reasonable set of features beyond the core feature listed in the table.

Deal Breakers

- None.

Other Considerations

- Customers in these markets have a slight preference for wearing devices on their wrists and dislike to wear them around their chests.
- Minor considerations such as the convenience of various power configurations, GPS connectivity, and the appeal of upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 30.

MD Heart Market

This market is comprised mainly of cardiovascular physicians who provide monitors to patients concerned with heart disease. The temporal monitor might be a convenient replacement for on-site evaluations that cost \$250-\$500.

Regional Market Size

Region	Market Size
Each Region	20,000-40,000

Bass Model Parameters

Parameter	Value
p	0.0002
incremental p from every \$500 in daily advertising	0.0002
q	0.0035

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Heartbeat)	WTP
Pulse only	0-115
Temporal	600-865

Note: These WTP values assume products that contain a reasonable set of features beyond one of the core features listed in the table.

Deal Breakers

- While not a complete deal breaker for all customers, lack of GPS can significantly drive down the perceived value of the product, since it's seen as a safety mechanism for dealing with patients who might experience an acute cardiovascular episode.

Other Considerations

- Minor considerations such as the convenience of various power configurations and the appeal of upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 30.

Product Design Guide

Different product features appeal in varying degrees to different market segments. Only one feature selection can be made for each product attribute. New products are developed as derivatives of existing products. In developing a derivative product, the company incurs design days and design cost only for incremental changes to the parent product's features. Costs are additive across features. The "None" option is only used to remove a feature from a parent product when developing a derivative product. If the parent product does not contain the feature, there is no cost to exclude it from the derivative.

Attribute	Feature	Design days	Design cost	Materials cost/unit
Heartbeat	None	3	1,000	0
	Pulse only	15	30,000	15
	Temporal	90	135,000	25
Blood vessel	None	3	1,000	0
	Systolic only	30	75,000	10
	Systolic & diastolic	90	135,000	15
	Full profile	120	180,000	40
Dissolved gasses	None	3	1,000	0
	O2 only	30	75,000	15
	O2, N2, CO2	90	135,000	20
	Full C, N, O	90	135,000	40
Toxicology	None	3	1,000	0
	Ethanol	30	150,000	95
	Amphetamine	90	250,000	140
	THC	90	250,000	140
	Barbiturate	90	250,000	140
	Narcotic	90	250,000	140
Hormone	None	3	1,000	0
	LH	30	45,000	20
	LH and FSH	60	75,000	50
	Estrogen	60	75,000	60
	Progesterone	60	75,000	60
	Testosterone	60	75,000	50
Metabolic	None	3	1,000	0
	Thyroxine	90	90,000	155
	Bilirubin	90	90,000	150
	Proteins	90	90,000	170
	Uric acid	90	90,000	160

Attribute	Feature	Design days	Design cost	Materials cost/unit
Cancer	None	3	1,000	0
	Base	60	200,000	100
	Prostate	90	300,000	210
	Breast	90	300,000	200
	Bladder and kidney	90	300,000	300
	Lymphoma	90	300,000	250
	Blood and bone	90	300,000	310
Neurotoxins	None	3	1,000	0
	Botulinum	90	135,000	190
	Anatoxin-a	90	135,000	210
	Sarin and cyclosarin	90	135,000	220
	Soman	90	135,000	280
Motion	None	3	1,000	0
	Steps	15	30,000	15
	Steps plus balance	30	45,000	30
	Steps plus balance & gait	45	60,000	45
Platform	Wrists	90	135,000	20
	Chest	15	3,000	10
	Sleeves	30	30,000	15
	Stockings	30	30,000	15
GPS	No GPS	3	1,000	0
	GPS	30	45,000	50
Network	Bluetooth	15	1,000	5
	2.4 GHz	30	30,000	10
	5 GHz	45	36,000	20
Power	Ni-Cd	5	1,500	5
	Ni-Cd pack	10	15,000	20
	Polymer	5	1,500	35
	Polymer pack	10	15,000	140
Finish	Original	3	2,400	0
	Blue	5	3,000	3
	Red	5	3,000	3
	Green	5	3,000	3
	Black	5	3,000	3
	White	5	3,000	3
	Metallic	90	27,000	6
	Geometric	90	27,000	6
	Camouflage	20	27,000	6